ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 C ✓ 1.1.5 B ✓ 1.1.6 A ✓ 1.1.7 A ✓ 1.1.8 B ✓ 1.1.9 D ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 E ✓ 1.3.3 G ✓ 1.3.4 B ✓ 1.3.5 I ✓ 1.3.6 J ✓ 1.3.7 A ✓ 1.3.8 F ✓ 1.3.9 D ✓ 1.3.10 H ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Combining mechanical engineering, electronics and computer control in one product or system ✓. Any TWO examples ✓✓: washing machine, robot vacuum, automatic gate, ATM, 3D printer, car ABS, drone. (3)

2.2 Any FOUR, each with its hazard ✓✓✓✓: safety goggles – flying chips and sparks; ear muffs – loud noise; safety boots – falling objects; gloves – sharp edges and heat; overall – loose clothing caught in machines; dust mask – dust and fumes. (4)

2.3 (a) side cutter / wire cutter ✓ (b) spanner / socket ✓ (c) micrometer ✓ (d) centre punch ✓ (4)

2.4 Any THREE ✓✓✓: guards must be in place; never leave a running machine; switch off and isolate before adjusting or cleaning; no loose clothing, jewellery or long hair; know where the emergency stop is; only use machines you are trained on. (3)

[14]

QUESTION 3

3.1.1 $\text{Clockwise moments} = \text{anticlockwise moments: } 50 \times 1{,}2 = \text{L} \times 0{,}3$ ✓✓; $\text{L} = 200\ \text{N}$ ✓ (3)

3.1.2 $\text{MA} = 200 \div 50 = 4$ ✓ (1)

3.1.3 $\text{VR} = 1{,}2 \div 0{,}3 = 4$ ✓ (1)

3.2 $\text{MA} = 600 \div 200 = 3$ ✓; $\text{VR} = 4 \div 1 = 4$ ✓; $\eta = \text{MA} \div \text{VR} \times 100$ ✓ $= 75\%$ ✓ (4)

3.3 First class ✓ – e.g. see-saw, pliers, crowbar ✓; second class ✓ – e.g. wheelbarrow, bottle opener ✓; third class ✓ – e.g. tweezers, fishing rod, human forearm ✓ (6)

3.4 Reason ✓: some energy is lost to friction (heat, noise) and lifting moving parts. Improvements (any TWO) ✓✓: lubrication; smooth bearings; lighter moving parts; good alignment. (3)

3.5 $\mathrm{W} = \mathrm{F} \times \mathrm{s}$ ✓ $= 600 \times 1 = 600\ \mathrm{J}$ ✓ (2)

[20]

QUESTION 4

4.1 $\mathrm{GR} = 45 \div 15 = 3:1$ ✓; $\mathrm{N} = 900 \div 3$ ✓ $= 300\ \mathrm{r/min}$ ✓ (3)

4.2 The gear ratio stays the same ✓; the driven gear now turns in the same direction as the driver ✓ (2)

4.3 $\mathrm{N}_2 = 1\,200 \times 80 \div 240$ ✓ $= 400\ \mathrm{r/min}$ ✓ (2)

4.4 Gears (any TWO) ✓✓: no slip; exact speed ratio; compact; carry large loads. Belts (any TWO) ✓✓: quiet; cheap; absorb shocks; can connect shafts far apart; slip protects against overload. (4)

4.5 Any FOUR with uses ✓✓✓✓: rack and pinion – steering, sliding gate; cam and follower – engine valves, toy movements; crank and slider – engines, pumps; screw and nut (lead screw) – vice, 3D-printer axis; lever – linkages; pulley – lifting. (4)

4.6 $\mathrm{T}_{\mathrm{out}} = 2 \times 3$ ✓ $= 6\ \mathrm{N{\cdot}m}$ ✓ (2)

4.7 To reduce friction and wear (and heat / noise) ✓ (1)

[18]

QUESTION 5

5.1 Material ✓ and reason ✓ each: (a) aluminium (or acrylic / PLA) – light, strong, does not rust; (b) copper – excellent conductor, flexible; (c) nylon / acetal plastic – quiet, low friction, cheap, self-lubricating; (d) PVC / rubber – good insulator, flexible. (8)

5.2 The Arduino pin (through a transistor) switches the small current in the relay coil ✓; the coil becomes an electromagnet and pulls the contacts closed ✓; the contacts switch the separate 12 V circuit of the motor, so the Arduino is not overloaded and is isolated from the motor ✓. (3)

5.3 Single-acting: air pushes the piston one way only ✓, and a spring returns it ✓. Double-acting: air pushes the piston both ways (out and in) ✓, through two ports, giving force in both directions ✓. (4)

5.4 Any THREE ✓✓✓: compressor; air receiver / tank; FRL unit (filter, regulator, lubricator); directional control valve; pipes and fittings; cylinder / actuator. (3)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE